

Llama Index vs LangChain

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Agenda:

1. What is llama Index?
2. What is Lang Chain?
3. What is difference between Llama Index and Lang Chain?
4. Where we should which one?
5. Can we combine both for our application building?

What is Llama Index?

Llama Index is an **AI data framework** that helps Large Language Models (LLMs) like GPT, Gemini, Llama, Mistral, etc. **connect to your own data.**

Think of it as a **bridge** between **your data** and an **LLM.**

Layman Analogy:

Imagine:

- LLM (ChatGPT/Gemini) = A **very intelligence** student.
- Your documents = Thousands of books in a **library.**
- LlamaIndex = The **librarian.**

Llama Index supported LLMs:

OpenAI (GPT), Google Gemini, Anthropic Claude, Meta Llama, Mistral, Cohere, Ollama (Local models), Hugging Face models, NVIDIA NIMs.

Major Components of Llama Index:

Component	Purpose
Readers	Load data from different sources
Documents	Store loaded text
Nodes	Small chunks of documents
Embeddings	Convert text into vectors
Index	Organize data for retrieval
Retriever	Find relevant chunks
Query Engine	Answer user questions
Chat Engine	Enable conversational interactions
Response Synthesizer	Combine retrieved context into a final answer

What is LangChain?

LangChain is an **open-source framework** for building applications powered by **Large Language Models (LLMs).**

If **LlamaIndex** is mainly responsible for **connecting LLMs to your data**, then **LangChain** is responsible for **orchestrating the entire AI application's workflow**.

Layman Analogy

Imagine you're building an AI travel assistant.

- **LLM (GPT/Gemini)** = A smart travel expert.
- **LlamaIndex** = The librarian who finds travel information.
- **LangChain** = The travel manager who coordinates everything.

The travel manager can:

- Ask the librarian for information.
- Check flight prices using an API.
- Book a hotel.
- Remember previous conversations.
- Calculate travel costs.
- Return one final response.

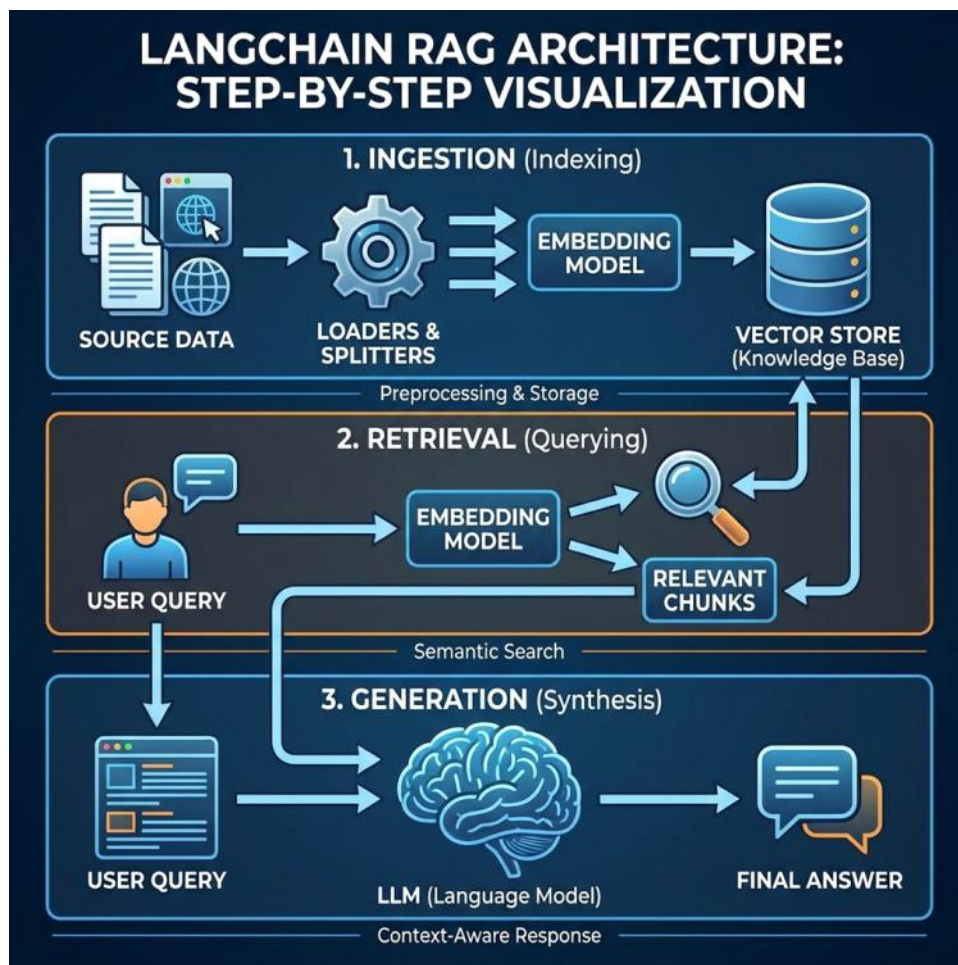
That "manager" is **LangChain**.

Component	Purpose	Example
Models	Connects to LLMs and embedding models	GPT-4, Gemini, Claude, Llama
Prompt Templates	Creates reusable prompts with	"Explain {topic} in simple
Chains (LCEL/Runnables)	Connects multiple steps into a	Prompt → LLM → Output
Agents	Lets the LLM decide which tool to	ReAct Agent, Tool Calling Agent
Tools	External functions the agent can	Calculator, Search API, Python
Retrievers	Retrieves relevant documents	Vector Store Retriever
Memory	Maintains conversation history	Chat History, Conversation
Document Loaders	Loads data from different sources	PDF, DOCX, CSV, Websites
Text Splitters	Splits large documents into chunks	RecursiveCharacterTextSplitter
Embeddings	Converts text into vectors	OpenAI Embeddings, BGE, Hugging Face
Vector Stores	Stores and searches embeddings	FAISS, Chroma, Pinecone,
Output Parsers	Converts LLM output into structured formats	JSON, Pydantic Models
Callbacks	Logs and monitors execution	Token usage, latency tracking
Chat Message History	Stores previous chat messages	Redis, SQL, File
Indexes (through retrievers/vector stores)	Enables efficient document search	Vector Index
RAG Pipeline	Combines retrieval with	Retrieve → Prompt → LLM
LangServe	Deploys LangChain applications as	FastAPI-based serving
LangSmith	Debugging, tracing, and evaluation platform	Prompt debugging, execution
Streaming	Streams responses token by token	Real-time chatbot responses
Multi-Agent Support	Multiple agents collaborate	Research Agent + Coding Agent

Llama Index vs Lang Chain:

Feature	Llama Index	Lang Chain
Primary focus	Data ingestion and retrieval	End-to-end LLM application orchestration
Best at	RAG & Docs querying	Agents, workflows, tools & chains
Learning curve	Generally simpler	Broader, often more complex
Data connectors	Excellent	Good
Agent capabilities	Basic to moderate	Extensive
Typical use case	Search your documents	Build complete AI assistants and workflows

They are not competitors so much as complementary tools. Many production systems use **LlamaIndex** for retrieval and **LangChain** for orchestration.



Llama Index:-

1. Data Ingestion
2. Search
3. Retrieval
4. Indexing
5. Ranking and Structuring

Points to be remember about Llama Index:

- LlamaIndex is specifically designed for **search, retrieval, and question-answering applications**.
- It acts as a bridge between your data and Large Language Models (LLMs).
- It provides powerful **data ingestion** capabilities for integrating custom data with LLMs.
- It supports **structured, semi-structured, and unstructured** data sources.
- It automatically performs document parsing, chunking, embedding generation, indexing, retrieval, and response synthesis.
- It integrates with many vector databases such as **FAISS, Chroma, Pinecone, Milvus, Qdrant, and Weaviate**.
- It works with multiple embedding models and LLM providers, including OpenAI, Google Gemini, Anthropic Claude, Meta Llama, Ollama, Hugging Face, and others.
- It is one of the most widely used frameworks for building **Retrieval-Augmented Generation (RAG)** applications.

Interview Definition:

Llama Index is an open-source data framework that connects LLM with private data by performing **data ingestion, Indexing, Semantic search, Retrieval, and Context preparation**, enabling accurate **Retrieval-Augmented Generation (RAG)** applications.

Use **Llama Index** when the **primary challenge is connecting an LLM** to your data through **ingestion, indexing, and retrieval**—for example, building a **RAG system** over **PDFs or databases**. Use **Lang Chain** when you need to **orchestrate LLM workflows**, such as **agents, tool calling, API integrations, memory, or multi-step automation**. In many production systems, they are used together: **Llama Index handles retrieval, while Lang Chain manages the overall application workflow**.

Easy way to remember

Framework	Think of it as	Main Job
Llama Index	Librarian	Finds the right information from your data
Lang Chain	Project Manager	Coordinates the entire workflow, tools, and decision-making